Assessment Menu: Teacher Guide

The Transformations Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
PLANE	Distance, midpoint, and slope	A1, A2, E1, S1
SHIFT	Translations and reflections	A3, A4, E2, S2
SPIN	Rotations and dilations	A5, E3, E4, S3
COMBINE	Compositions and describing transformations	A6, E5, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 PLANE	Find the distance between A(1, 2) and B(4, 6).	**Answer:** AB = 5 $d = \sqrt{((4-1)^2 + (6-2)^2)} = \sqrt{(9 + 16)}$ $d = \sqrt{25} = 5$	3 pts: Correct distance formula with substitution 2 pts: AB = 5
A2 PLANE	Find the midpoint and the slope of the segment with endpoints A(−2, 5) and B(6, −1).	**Answer:** Midpoint (2, 2); slope $\frac{-3}{4}$ Midpoint = $(\frac{-2+6}{2}, \frac{5+(-1)}{2})$ = (2, 2) Slope = $\frac{-1-5}{6-(-2)} = \frac{-6}{8} = \frac{-3}{4}$	3 pts: Midpoint (2, 2) 2 pts: Slope $\frac{-3}{4}$
A3 SHIFT	Triangle PQR has P(−3, 2), Q(1, 4), R(0, −1). Translate it using (x, y) → (x + 4, y − 3). Graph the image and label P′, Q′, R′.	**Answer:** P′(1, −1), Q′(5, 1), R′(4, −4) Add 4 to each x and subtract 3 from each y: P′(−3 + 4, 2 − 3) = (1, −1) Q′(1 + 4, 4 − 3) = (5, 1) R′(0 + 4, −1 − 3) = (4, −4). Plot and connect.	3 pts: Three correct image points (1 pt each) 2 pts: Plots and connects the image triangle, labeled
A4 SHIFT	Point A is at (3, −5). Give the coordinates of its reflection across the x-axis and its reflection across the y-axis.	**Answer:** Across x-axis: (3, 5); across y-axis: (−3, −5) Reflecting across the x-axis changes the sign of y: (3, 5). Reflecting across the y-axis changes the sign of x: (−3, −5).	2 pts: Reflection across the x-axis (3, 5) 3 pts: Reflection across the y-axis (−3, −5)
A5 SPIN	Rotate the point (2, 5) by 90° counterclockwise about the origin. Use the rule (x, y) → (−y, x).	**Answer:** (−5, 2) (x, y) → (−y, x), so (2, 5) → (−5, 2).	2 pts: Uses the rule (−y, x) 3 pts: Correct image (−5, 2)
A6 COMBINE	Point (−2, 3) is reflected across the y-axis, then translated 5 units down. Find its final image.	**Answer:** (2, −2) Reflect across the y-axis: (−2, 3) → (2, 3). Translate 5 units down: (2, 3 − 5) = (2, −2).	2 pts: After the reflection (2, 3) 1 pts: Moves down by subtracting from y 2 pts: Final image (2, −2)

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 PLANE	Triangle ABC has A(−1, 1), B(3, 4), C(6, 0). **(a)** Use the distance formula to find AB, BC and AC. **(b)** Use slopes to decide whether ∠B is a right angle. **(c)** Classify the triangle by its sides.	**Answer:** (a) AB = 5, BC = 5, AC = $5\sqrt{2}$ (b) Yes, slopes $\frac{3}{4}$ and $\frac{-4}{3}$ are perpendicular (c) Isosceles (a) AB = $\sqrt{(4^2 + 3^2)} = 5$; BC = $\sqrt{(3^2 + (-4)^2)} = 5$; AC = $\sqrt{(7^2 + (-1)^2)} = \sqrt{50} = 5\sqrt{2}$ (b) Slope AB = $\frac{3}{4}$; slope BC = $\frac{-4}{3}$. The product is −1, so AB ⊥ BC and ∠B is a right angle. (c) AB = BC, so the triangle is isosceles (a right isosceles triangle).	3 pts: Three distances (AC simplified to $5\sqrt{2}$) 3 pts: Both slopes correct 2 pts: Product of slopes is −1, so perpendicular 2 pts: Classifies as isosceles using AB = BC
E2 SHIFT	A translation maps A(−2, 3) to A′(4, −1). **(a)** Write the translation rule. **(b)** Find the image of B(1, −5). **(c)** Which point is mapped to (0, 0)?	**Answer:** (a) (x, y) → (x + 6, y − 4) (b) B′(7, −9) (c) (−6, 4) (a) x: −2 → 4 is +6. y: 3 → −1 is −4. Rule: (x + 6, y − 4). (b) B′ = (1 + 6, −5 − 4) = (7, −9) (c) Work backward: (0 − 6, 0 + 4) = (−6, 4). Check: (−6 + 6, 4 − 4) = (0, 0).	3 pts: (a) Rule (x + 6, y − 4) 3 pts: (b) B′(7, −9) 4 pts: (c) Works backward (undoes the translation) to (−6, 4)
E3 SPIN	Triangle ABC has A(2, −4), B(6, −2), C(4, 4). Dilate it with center at the origin and scale factor $\frac{1}{2}$. Graph both triangles, list the image points, and compare AB with A′B′ using the distance formula.	**Answer:** A′(1, −2), B′(3, −1), C′(2, 2); AB = $2\sqrt{5}$, A′B′ = $\sqrt{5}$, ratio $\frac{1}{2}$ Multiply each coordinate by $\frac{1}{2}$: A′(1, −2), B′(3, −1), C′(2, 2). AB = $\sqrt{(4^2 + 2^2)} = \sqrt{20} = 2\sqrt{5}$ A′B′ = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$ Ratio $\frac{\sqrt{5}}{2\sqrt{5}} = \frac{1}{2}$, equal to the scale factor.	3 pts: Three correct image points 2 pts: Graphs both triangles 3 pts: AB = $2\sqrt{5}$ and A′B′ = $\sqrt{5}$ 2 pts: Ratio equals the scale factor $\frac{1}{2}$
E4 SPIN	Triangle DEF has D(1, 2), E(4, 2), F(4, 5). Rotate it 90° clockwise about the origin using (x, y) → (y, −x). Graph the image and use the distance formula to compare DE with D′E′. What does this show?	**Answer:** D′(2, −1), E′(2, −4), F′(5, −4); DE = D′E′ = 3, so rotations preserve length. (x, y) → (y, −x): D′(2, −1), E′(2, −4), F′(5, −4). DE = $\sqrt{((4 - 1)^2 + 0^2)} = 3$ D′E′ = $\sqrt{(0^2 + (-4 - (-1))^2)} = 3$ The lengths are equal, so a rotation preserves distance (it is a rigid motion).	3 pts: Three correct image points 2 pts: Graphs the image triangle 3 pts: DE = 3 and D′E′ = 3 by the distance formula 2 pts: Concludes rotations preserve length
E5 COMBINE	P(2, 5) is reflected across the line y = x, then translated 3 left and 1 up. Lena wrote: “Reflection: (2, 5) → (−5, −2). Translation: (−8, −1).” Find her mistake and give the correct final image.	**Answer:** Reflecting across y = x swaps the coordinates (no sign change). Correct final image: (2, 3). Reflection across y = x swaps x and y: (2, 5) → (5, 2). Lena also changed both signs, which is a reflection across y = −x. Translate 3 left and 1 up: (5 − 3, 2 + 1) = (2, 3).	4 pts: Identifies mistake: y = x just swaps the coordinates 3 pts: Correct reflection (5, 2) 3 pts: Correct final image (2, 3)

E6 COMBINE	Triangle ABC with A(1, 1), B(4, 1), C(4, 3) is mapped to A′(−1, −1), B′(−4, −1), C′(−4, −3). **(a)** Describe the single transformation and write its rule. **(b)** Show that the origin is the midpoint of AA′.	**Answer:** (a) Rotation of 180° about the origin, (x, y) → (−x, −y) (b) Midpoint of AA′ = (0, 0) (a) Every coordinate changes sign, so (x, y) → (−x, −y): a 180° rotation about the origin. (b) Midpoint of AA′ = $\left(\frac{1 + (-1)}{2}, \frac{1 + (-1)}{2}\right)$ = (0, 0).	4 pts: Names a 180° rotation about the origin 3 pts: Rule (x, y) → (−x, −y) 3 pts: Midpoint computed to be (0, 0)

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 PLANE	A(−2, 1), B(1, 5), C(7, 3) are three vertices of parallelogram ABCD, in that order. **(a)** Use the fact that diagonals bisect each other to find D. **(b)** Use slopes to verify ABCD is a parallelogram. **(c)** Is it a rectangle? Justify with slopes.	**Answer:** (a) D(4, −1) (b) Opposite sides parallel (c) Not a rectangle (a) Midpoint of AC = $(\frac{5}{2}, 2)$ = midpoint of BD, so D = (4, −1). (b) Slope AB = slope DC = $\frac{4}{3}$; slope BC = slope AD = $\frac{-1}{3}$. Opposite sides are parallel. (c) Slopes of AB and BC multiply to $\frac{-4}{9}$, not −1, so ∠B is not a right angle: not a rectangle.	3 pts: (a) Midpoint of AC is $(\frac{5}{2}, 2)$ 3 pts: (a) D = (4, −1) 4 pts: (b) All four slopes correct 2 pts: (b) Concludes both pairs of opposite sides are parallel 4 pts: (c) Product of adjacent slopes is not −1 4 pts: (c) Concludes not a rectangle
S2 SHIFT	Triangle LMN has L(2, 1), M(5, 1), N(5, 5). **(a)** Reflect it across the line x = 1. Graph both triangles and the line, and list L′, M′, N′. **(b)** Show the line x = 1 is the perpendicular bisector of MM′ using the midpoint and slopes.	**Answer:** (a) L′(0, 1), M′(−3, 1), N′(−3, 5) (b) Midpoint of MM′ is (1, 1), on the line; MM′ is horizontal, the line vertical. (a) Each point goes the same distance to the other side of x = 1. (b) Midpoint of MM′ = $(\frac{5 + (-3)}{2}, 1)$ = (1, 1), on x = 1. MM′ is horizontal and x = 1 is vertical, so they are perpendicular: x = 1 is the perpendicular bisector.	6 pts: (a) Three correct image points (2 pts each) 4 pts: (a) Graphs the line, triangle, image and labels 4 pts: (b) Midpoint (1, 1) lies on the line 4 pts: (b) Slope reasoning: horizontal segment, vertical line 2 pts: (b) Concludes perpendicular bisector
S3 SPIN	A dilation centered at the origin maps A(3, −6) to A′(1, −2). **(a)** Find the scale factor. **(b)** Find the image of B(−9, 12). **(c)** Find AB and A′B′ in simplest radical form and show their ratio equals the scale factor. **(d)** Triangle ABC has area 54. Find the area of its image and explain.	**Answer:** (a) $\frac{1}{3}$ (b) B′(−3, 4) (c) $6\sqrt{13}$, $2\sqrt{13}$ (d) 6 (a) k = $\frac{1}{3}$ (since $1 = \frac{1}{3} \cdot 3$). (b) B′ = (−3, 4) (c) AB = $\sqrt{468} = 6\sqrt{13}$; A′B′ = $\sqrt{52} = 2\sqrt{13}$; ratio $\frac{1}{3}$. (d) Areas scale by k^2: $54 \cdot \frac{1}{9} = 6$.	4 pts: (a) Scale factor $\frac{1}{3}$ 4 pts: (b) B′(−3, 4) 6 pts: (c) AB = $6\sqrt{13}$ and A′B′ = $2\sqrt{13}$ (2 pts each), ratio $\frac{1}{3}$ (2 pts) 3 pts: (d) Area 6 3 pts: (d) Explains areas scale by k^2 (the square of the scale factor)
S4 COMBINE	Triangle ABC has A(1, 2), B(5, 2), C(5, 4). **(a)** Rotate it 90° counterclockwise about the origin using (x, y) → (−y, x). **(b)** Reflect that image across the x-axis. Graph ABC and the final triangle and list the final points. **(c)** Name the single transformation with the same result. **(d)** If A is reflected first and rotated second, does it land at the same point? Explain.	**Answer:** (a) A′(−2, 1), B′(−2, 5), C′(−4, 5) (b) A″(−2, −1), B″(−2, −5), C″(−4, −5) (c) Reflection across y = −x (d) No: (2, 1) (a) (x, y) → (−y, x) gives the points in (a). (b) Negate y to get the points in (b). (c) Overall (x, y) → (−y, −x), a reflection across y = −x. Check: A(1, 2) → (−2, −1). (d) Reflect first: (1, 2) → (1, −2) → (2, 1), not (−2, −1). Order matters.	3 pts: (a) Rotation points 3 pts: (b) Final points 4 pts: Graphs ABC and A″B″C″ 5 pts: (c) Reflection across y = −x with rule or check 5 pts: (d) A lands at (2, 1), not (−2, −1); order matters